Use-Case Flow: Request Emergency Blood

Problem Statement #12 — Blood Bank Inventory & Emergency Donor Matcher

Actor: Emergency Requester (primary), Donor and Blood Bank Manager (secondary)

Preconditions: The Emergency Requester is authenticated in the system, and the system has live access to the blood inventory and donor databases.

Postconditions: Compatible donors have been identified and notified (or stock has been reserved directly), and the request status is logged as "Pending" or "Fulfilled".

Main Success Scenario

1. Emergency Requester selects “Request Emergency Blood.”
2. System displays a form for blood group, component type, quantity, and hospital location.
3. Requester submits the form.
4. System validates the input.
5. System checks current inventory for an immediate match.
6. If stock is insufficient, system searches the donor database within a 10 km radius («include» Search Compatible Donors).
7. System compiles the list of compatible, available donors.
8. System sends an SMS alert to each donor («include» Notify Donors via SMS).
9. System logs the notification event with a timestamp.
10. Use case ends with the request status set to “Pending Response.”

Alternate Flow — 8a. No donors found within 10 km

8a1. System triggers «extend» Escalate Search Radius, widening the search to 20 km.

8a2. System re-notifies the expanded donor list.

8a3. If no compatible donor responds within 15 minutes, the system flags the request as “Critical – Unfulfilled” and alerts the Blood Bank Manager.

Blood Bank Inventory & Emergency Donor Matcher

Requirements

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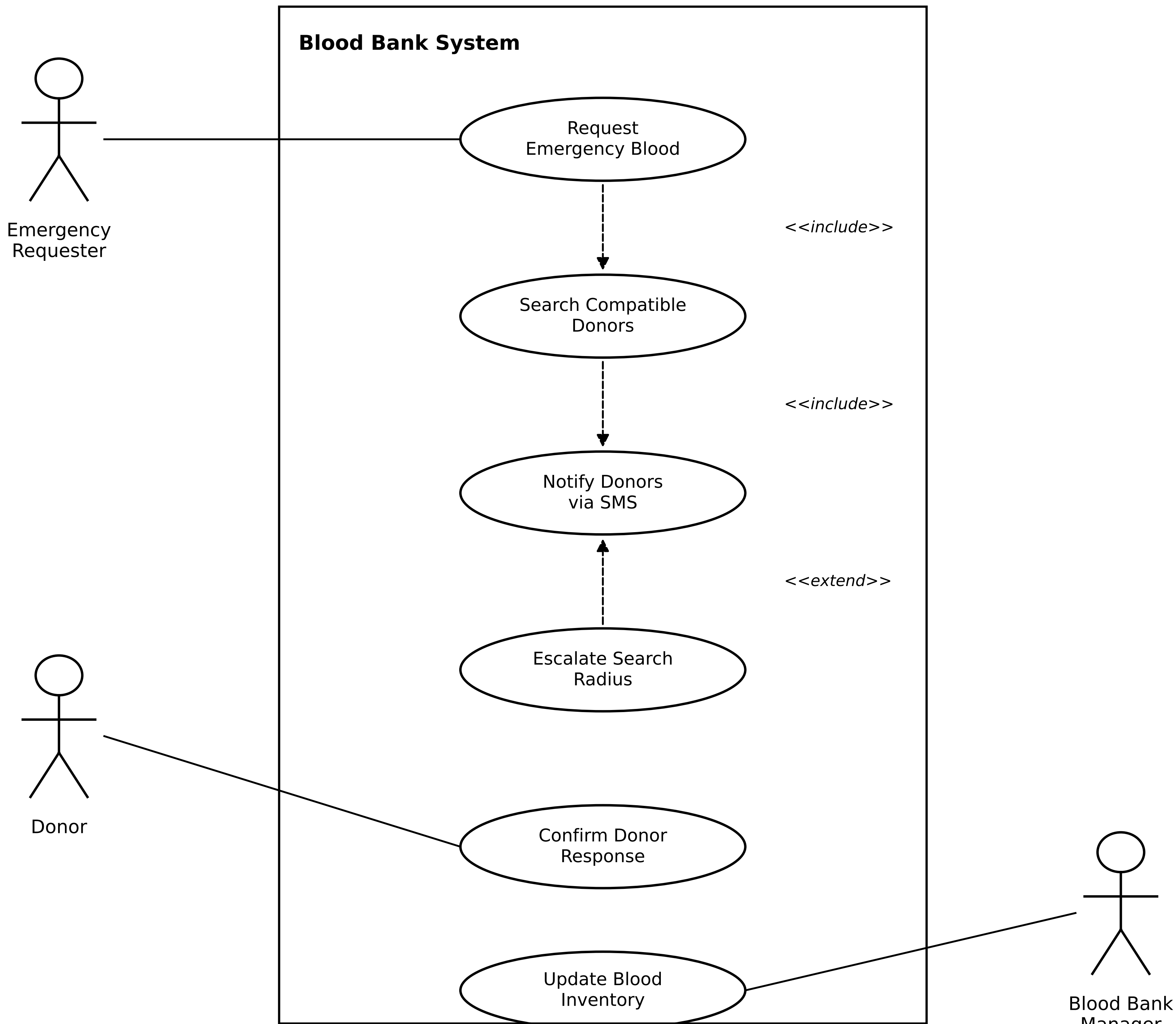
Req ID	Type	Description	Priority
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High
FR-002	Functional	The system shall allow an Emergency Requester to submit a request specifying blood group, component type, quantity, and hospital location.	High
FR-003	Functional	The system shall flag blood units nearing expiry (within 48 hours) for priority allocation or safe disposal.	Medium
FR-004	Functional	The system shall allow the Blood Bank Manager to view and update real-time stock levels per blood group and component type.	High
FR-005	Functional	The system shall record each donor's response (accept/decline) to a notification and update that donor's availability status accordingly.	Medium
NFR-001	Nonfunctional (Performance & Security)	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High
NFR-002	Nonfunctional (Security/Privacy)	Donor personal data (phone number, location) shall be encrypted in transit and at rest, with access restricted to authorized blood bank staff only.	High

Acceptance Criteria	Rationale
Pass: compatible donors notified within 30s of emergency flag. Fail: incompatible blood group alerted.	Core emergency-response function *(given)*
Pass: request is accepted only when all four fields are valid; invalid submissions are rejected with an inline error.	Ensures the system has enough data to match donors and stock accurately
Pass: a unit crossing the 48-hour threshold is auto-tagged "priority" in the inventory view within 5 minutes.	Reduces wastage and prevents use of expired units
Pass: a manual stock update reflects in the searchable inventory within 10 seconds.	Keeps the matching engine accurate
Pass: a donor who accepts is marked "committed" and excluded from further alerts for that request.	Prevents duplicate donor contact and double-commitment
Pass: benchmark/load tests confirm no double-allocation under concurrent requests; target latency met.	Prevents critical allocation errors *(given)*
Pass: security audit confirms encryption at rest/in transit and role-based access control; Fail: any unauthorized read access.	Protects sensitive donor data and supports compliance with health-data privacy norms


Blood Bank System
Emergency Requester
Request Emergency Blood
<<include>>
Search Compatible Donors
<<include>>
Notify Donors via SMS
<<extend>>
Escalate Search Radius
Donor
Confirm Donor Response
Update Blood Inventory
Blood Bank Manager